Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/09/2006
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Behavioral And Neurochemical Effects Of Exposure To Heavy Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	06/15/2003	**End Date:**	06/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Project extended to 06/30/2007 per PI (12/06) Received NCE thru 12/31/2006, per J. Dardano (2/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Joseph, James A (Human Nutrition Research Center on Aging, USDA) Shukitt-Hale, Barbara (Human Nutrition Research Center on Aging, USDA)		
Grant/Contract No.:	NAG9-1529		
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Performance Goal Text:			

Task Description:	Exposing rats to non-lethal doses of 56Fe particles produces changes in the central nervous system, particularly in dopaminergic function. Dopamine is a neurotransmitter that mediates a wide range of behaviors, including motor behavior, some types of learning, reinforcement, sensory information processing, food intake and the production of abnormal behavior patterns. Because changes in each of these behaviors can affect the performance of astronauts, the knowledge of the behaviors affected by exposure and an understanding of the neural mechanisms mediating the behaviors is essential. The proposed research program is designed to gain additional information on alterations in brain neurochemistry and behavior following exposure to particles of high energy and charge, such as those that might be encountered during space travel outside the magnetic field of the earth. The research is designed to further describe and characterize behavioral and neurochemical deficits produced by exposure to heavy particles, determine their underlying mechanisms, and develop approaches to minimize such deficits. In addition, the research program proposes to examine factors such as age and diet that may interact with the effects of exposure to heavy particles, and which may, in turn, affect the health and performance of astronauts during exploratory missions to other planets. Future missions in space (such as a mission to Mars) may involve long-term travel beyond the magnetic field of the earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. The objectives of the research project were to describe and characterize heavy particle-induced behavioral and neurochemical deficits, determine their underlying causes, and develop approaches to minimize such deficits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research which we have conducted previously has shown that exposing young rats (less than 3 mo. old) to low doses of heavy particles (56Fe, 600 MeV/n) has shown that the neurochemical deficits produced by this exposure are similar to those that are observed in aged rats (24 mo. old). The research which we have completed at BNL indicates that similar deficits are observed following exposure to one GeV/n 56Fe particles as well. Thus, exposing rats to low doses of 56Fe particles provides a way to accelerate aging in experimental animals with respect to certain brain and behavioral parameters so that experimenters do no have to wait for 24 months to obtain old animals. The research program is designed to understand the mechanisms by which exposure to heavy particles (primarily 56Fe) produce their effects on brain and behavior. Although the impetus for the research is to understand and minimize the effects of exposure to heavy particles on astronauts, the research program necessarily has implications for the understanding of the natural aging process. Because exposure to 56Fe particles may produce accelerated aging, this research is also indirectly concerned with the basic processes underlying the biology of aging. Similarly, because the ultimate goal of the research program is to develop interventions to minimize the effects of exposure to heavy particles on astronauts, the interventions may also prove useful with the natural aging process.
Task Progress:	Effects of Different Particles and Energies: The neurobehavioral effects of exposure to different particles and energies have yet to be established. Results indicate that in comparing different energies of 56Fe particles (600, 1000 and 5000 MeV/n; LET˜ 181, 150, 142 keV/µm, respectively) the effectiveness of the particle, defined as the threshold dose at which a performance decrement was observed, paralleled particle LET. Threshold dose for disruption of performance was lowest for 600 MeV/n 56Fe particles, highest for 5000 MeV/n 56Fe, and intermediate for the 1000 MeV/n 56Fe particles. A similar LET-dependent relationship was observed following exposure to 28Si particles (600 and 1000 MeV/n). However, when comparing the effectiveness of 28Si particles with that of 56Fe particles, exposure to 28Si particles produced behavioral deficits at lower doses than exposure to 56Fe particles, despite the lower LET of the 28Si particles. These results suggest that greater fluence of the lower LET 28Si particles may be a factor influencing the relative effectiveness of these particles in disrupting cognitive/behavioral performance, and that the overall effectiveness of different particles is determined by both the specific particle and the specific energy of that particle. Consistent with this hypothesis was the observation that the threshold for the disruption of neurobehavioral endpoints following exposure to 48Ti particles (1100 MeV/n, LET ˜ 106 keV/µm) was intermediate between that of 56Fe and 28Si partilces. As such it may not be possible to predict the effects of exposure to the range of HZE particles on the neurocognitive performance of astronauts on exploratory class missions based upon an examination of the effects of exposure to only a single particle. Effects of Diet on the Radiation-Induced Disruption of Behavioral and Neural Functioning: Previous research had shown that rats maintained on diets containing antioxidants such as 2% blueberry and strawberry extract for two months prior to exposure to 56Fe particles fail to show an irradiation-induced disruption of CTA learning produced by injection of amphetamine. The rats were maintained on this diet for a period of two months because this time frame had been previously shown to be effective in ameliorating the behavioral and neural changes associated with aging. However, in contrast to aging which is a long term process, exposing rats to heavy particles occurs within a very restricted time frame. As such we hypothesized that maintaining rats on diets containing strawberry or blueberry extract for two weeks would be effective in ameliorating the effects of exposure to 56Fe particles. In general, the results did not support the hypothesis that shorter term maintenance on antioxidant diets containing 2% or 4% strawberry or blueberry extract would ameliorate the behavioral deficits observed following exposure to 56Fe particles. In contrast to the results obtained following the two-month maintenance regimen, the rats maintained on these diets for two weeks continued to show decrements in performance when tested shortly following irradiation. These results suggest that a longer period of maintenance on these diets may be required to build up a sufficient reserve of free radical scavengers to prevent the disruption of these responses by HZE particle irradiation. Effects of Age and Diet on Behavioral Performance: Previous research has shown that there are a series of age-related decrements in performance following exposure to 56Fe particles. Operant responding on an ascending fixed-ratio schedule has been used to evaluate age-related decrements in performance because the same animals can be repeatedly tested on that task over intervals of up to 12 to 18 months following exposure to HZE particles. Because performance on this task deteriorates as a function of both age and irradiation, rats were retested on this task one year following exposure to 150 cGy of 56Fe particles. The results showed that the rats maintained on the 4%, but not the 2% antioxidant diets, showed no deficit in response rate compared to the non-irradiated rats maintained on the control diet. While these results are consistent with an interaction between age and the effects of exposure to HZE particles, they do not provide unequivocal support for a the short term use of antioxidant diets to provide protection against the deleterious combined effects of aging and HZE particle radiation of a variety of behavioral endpoints.
Bibliography Type:	Description: (Last Updated: 05/08/2025)

Articles in Peer-reviewed Journals	Rabin BM, Joseph JA, Shukitt-Hale B. "A longitudinal study of operant responding in rats irradiated when 2 months old." Radiat Res. 2005 Oct;164(4 Pt 2):552-5. PMID: 16187786 , Oct-2005
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Articles in Peer-reviewed Journals	Shukitt-Hale, B., Carey, A. N., Jenkins, D., Rabin, B. M., and Joseph, J. A. "Beneficial effects of fruit extracts on neuronal function and behavior in a rodent model of accelerated aging." Neurobiol Aging. in press, 2006. , Jun-2006
Significant Media Coverage	Giles J. "Article called “Space Anxieties.”" Newspaper interview for Society for Neuroscience. News@nature.com, Nov-2005